EECS 498 · AASE · FALL 2026

Browser Extensions and Assistants You Can Text

LECTURE 13 · OCT 13, 2026

Today's schedule

- **Part 1: one extension, many harnesses**
- **Demo**
- **Break**
- **Part 2: assistants you can text**
- **Questions**

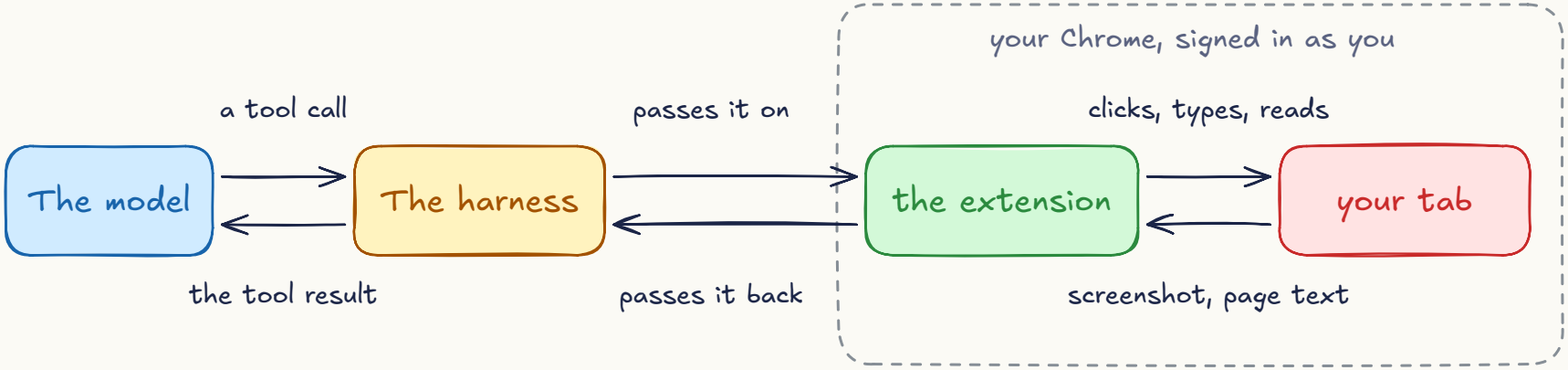
Lab04's browser part is the background: the three views of a page, and clicking by reference.

Part 1: One extension, many harnesses

WHO CAN DRIVE YOUR CHROME, AND HOW THEY REACH IT

One tool call, from the model to your tab

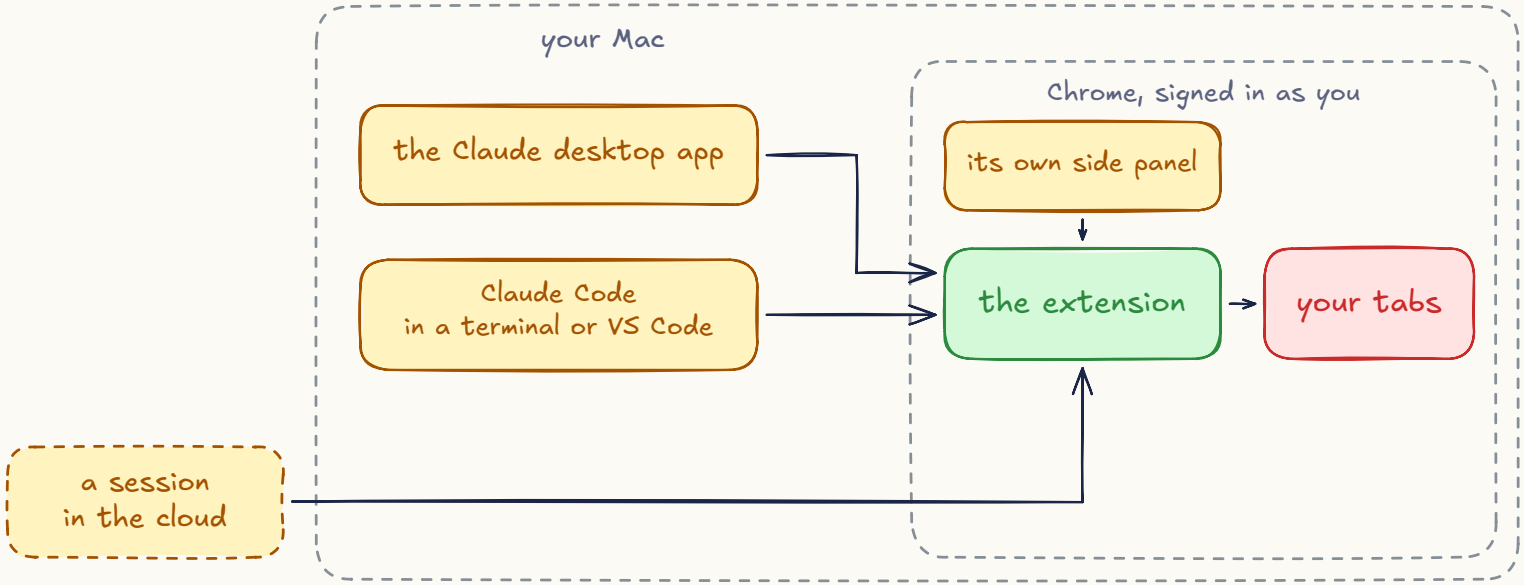
The harness passes each browser tool call to an extension inside your Chrome. The extension acts on the tab and sends the result back the same way.



Simplified from Claude Code 2.1.296 and the Claude in Chrome extension 1.0.100, read Oct 13. code.claude.com/docs/en/chrome.

Who can drive the extension

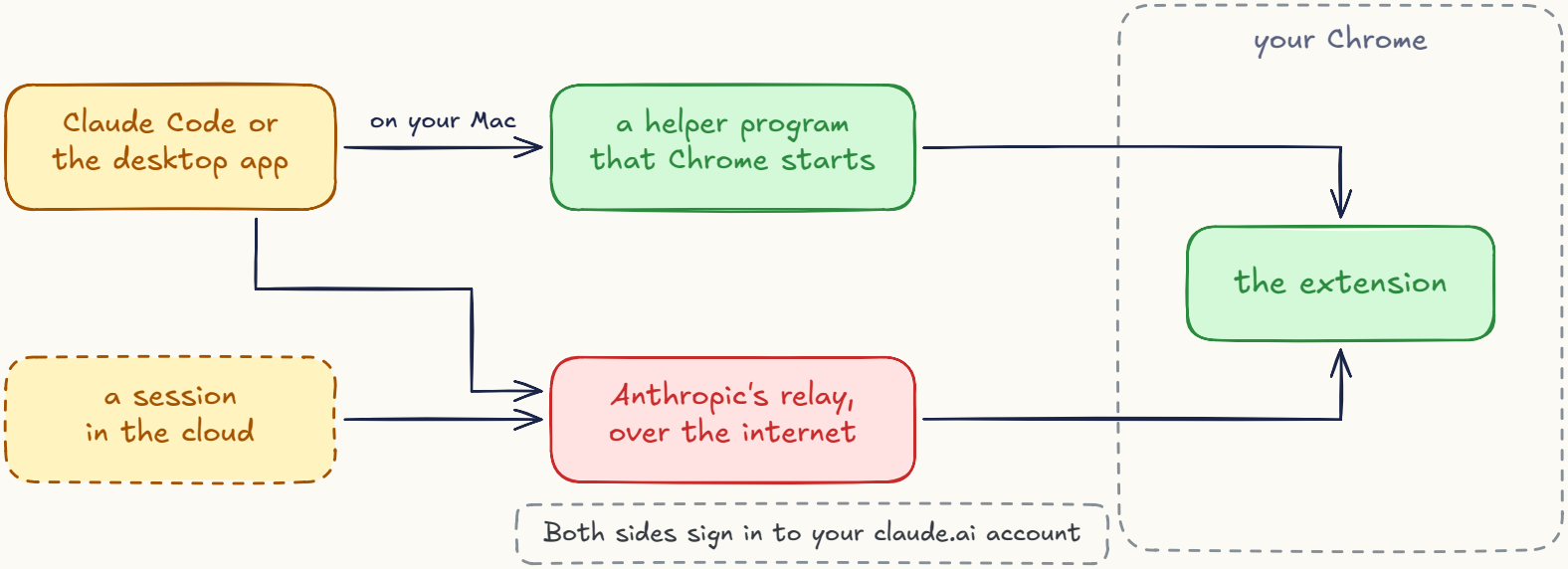
One extension in your Chrome takes tool calls from its own side panel, the desktop app, Claude Code, and even a session in the cloud.



code.claude.com/docs/en/chrome · the native messaging files for the desktop app and Claude Code on the instructor's Mac, Oct 13.

Two ways to reach the extension

A helper program that Chrome starts on your Mac, or Anthropic's relay over the internet. A cloud session can only use the relay.



Simplified from what the instructor's Mac showed on Oct 13 (Claude Code 2.1.296, extension 1.0.100) and code.claude.com/docs/en/chrome.

What the extension may do

Its manifest asks for the DevTools Protocol on every site.

Permission	What it allows
`debugger`	The DevTools Protocol on any tab: click, type, screenshot, run JavaScript
`<all_urls>`	Every site, including the ones you are signed into
`scripting`	Its own scripts inside pages
`tabs`, `tabGroups`	Open tabs and keep them in one group

manifest.json of Claude in Chrome 1.0.100, read Oct 13, 16 permissions in all. While it is attached, Chrome shows its own banner: "$1" started debugging this browser.

An extension with your logins

It can reach every site you are signed into. Four things hold it back.

Your logins

It shares your browser's sign-ins: email, Canvas, your bank

Its own tabs

The tabs it opens go into one group for the session

Site by site

It asks before each new site. Some kinds are blocked

It stops for you

On a login page or a CAPTCHA, it waits for you

code.claude.com/docs/en/chrome · support.claude.com/en/articles/12902428: adult and pirated sites blocked, financial sites ask first.

The same idea in Codex

Codex has its own Chrome extension, reached the first way: a helper on your Mac.

	Claude in Chrome	Codex's Chrome extension
Driven by	Its side panel, the desktop app, Claude Code, cloud sessions	Its side panel, the Codex app and CLI
Reached through	A helper on your Mac, or Anthropic's relay	A helper on your Mac
The model writes	One tool call per action	A short JavaScript program
Also asks for	None of these	Your history, bookmarks and sessions

Read Oct 13 on the instructor's Mac: Claude in Chrome 1.0.100 and OpenAI's ChatGPT extension 1.26.901.11451 · learn.chatgpt.com/docs/chrome-extension.

Demo

Chat break

What would you use browser automation for?

What can your harness do with a browser that it could not do before?

What browser automation is for

Three kinds of job: check what you built, reach what has no API, and repeat it without the model.

Check what you built

Open the page you just changed, read the console, try the form

Reach what has no API

Any site you are signed into: pull data out, fill a form in

Repeat it without the model

Turn one run into a test that runs on every push

The uses code.claude.com/docs/en/chrome lists: live debugging, design checks, web app testing, signed-in apps, data extraction, task automation · playwright.dev/docs/test-agents.

Part 2: Assistants you can text

WHY A HARNESS THAT IS ALWAYS ON

Where a session ends

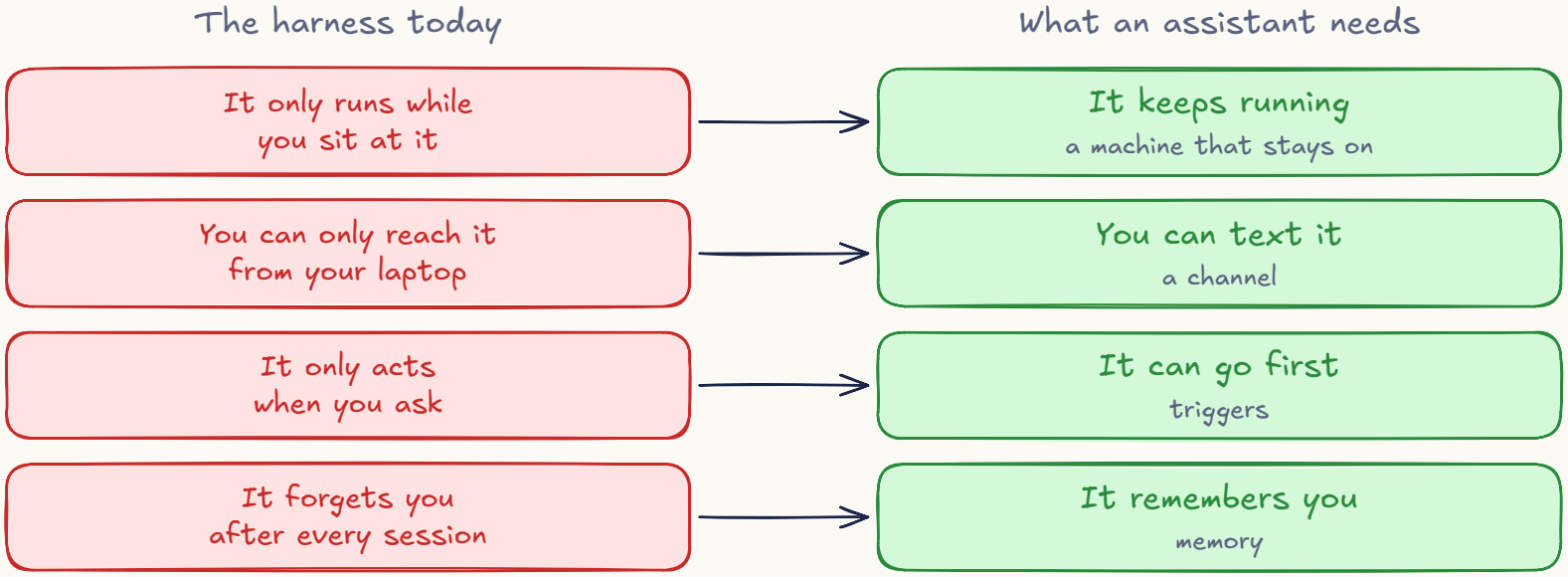
A harness only exists while your session is open. Most of what you want help with happens when it is not.



An illustrative day.

What an assistant needs

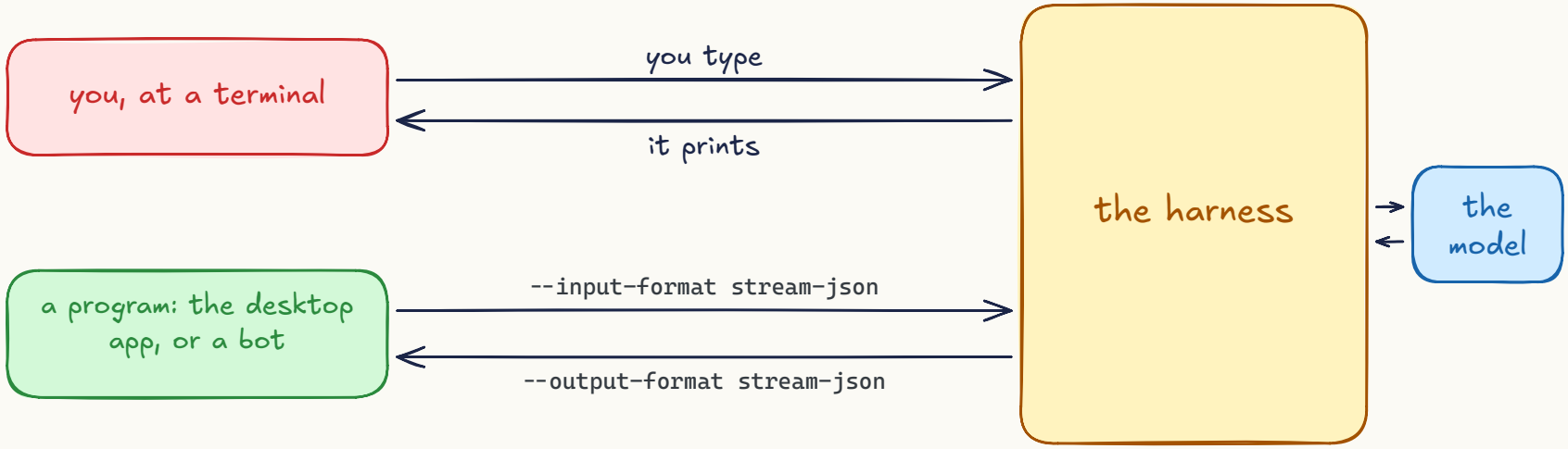
Four things the harness cannot do today, and the piece that fixes each one.



The four pieces are the ones OpenClaw, Hermes Agent and Claude Code with channels all build.

Driving the harness from a program

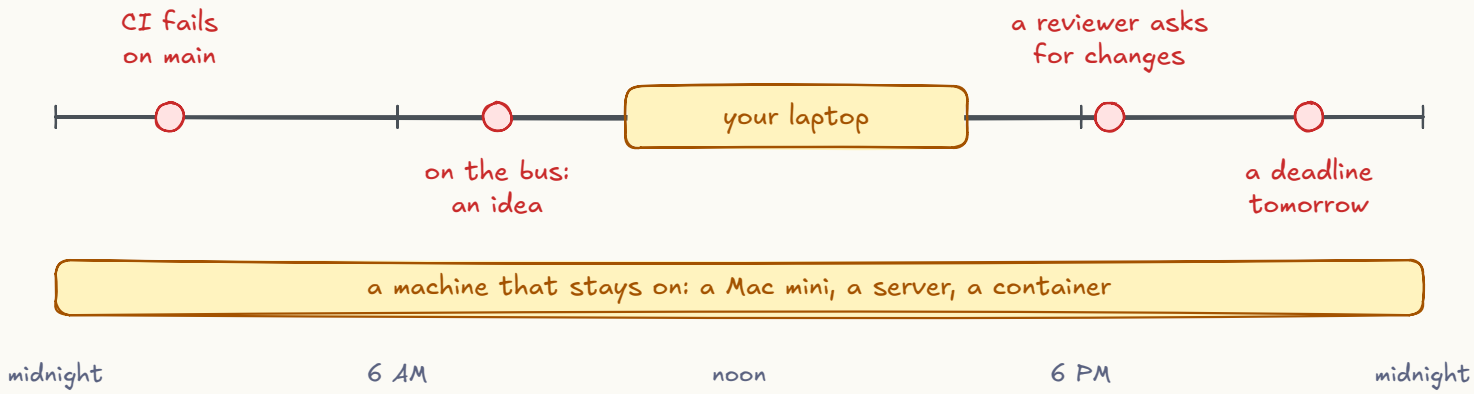
The harness can take its messages from a program instead of a person, and send its answers back as JSON.



Claude Code's stream-json flags. The Claude desktop app starts claude with both, per ps on the instructor's Mac, Oct 13. Codex has codex exec --json.

A machine that stays on

Put the harness on a machine that never closes, and every event in the day falls inside it.



The same illustrative day.

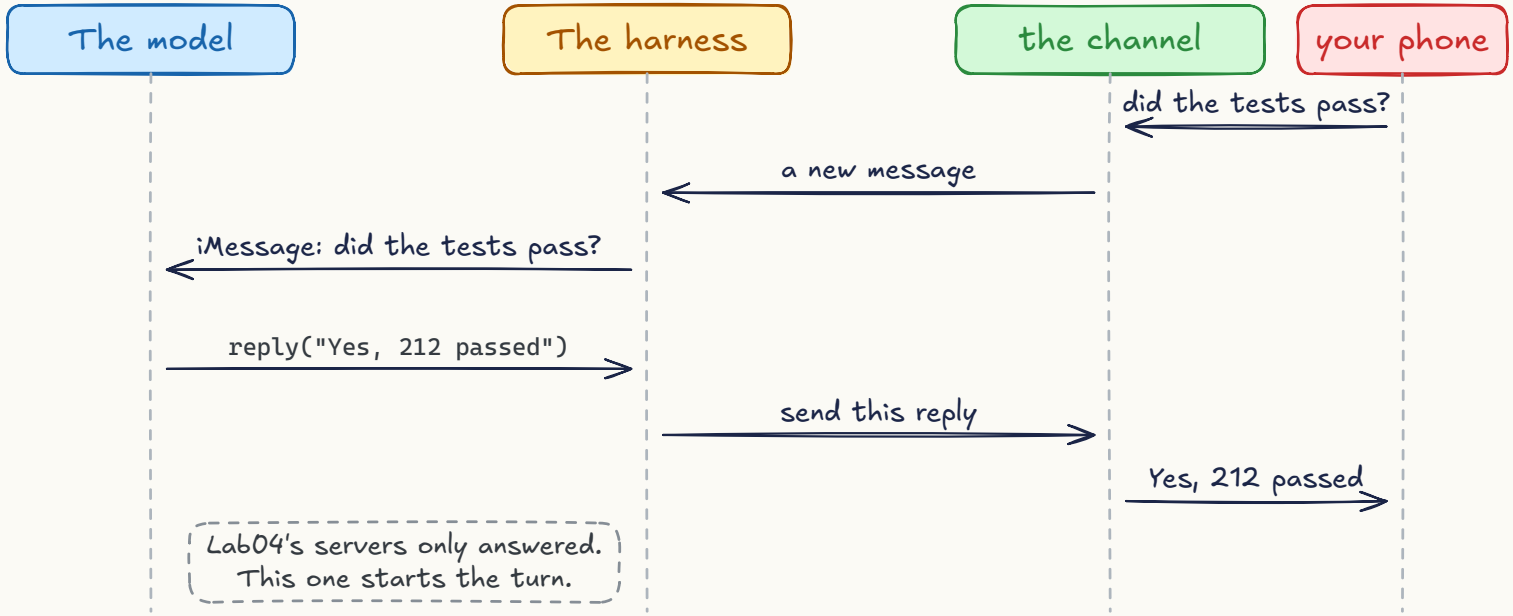
Machines that stay on, at home



The instructor's desk at home, Oct 2026.

A channel is an MCP server that starts the turn

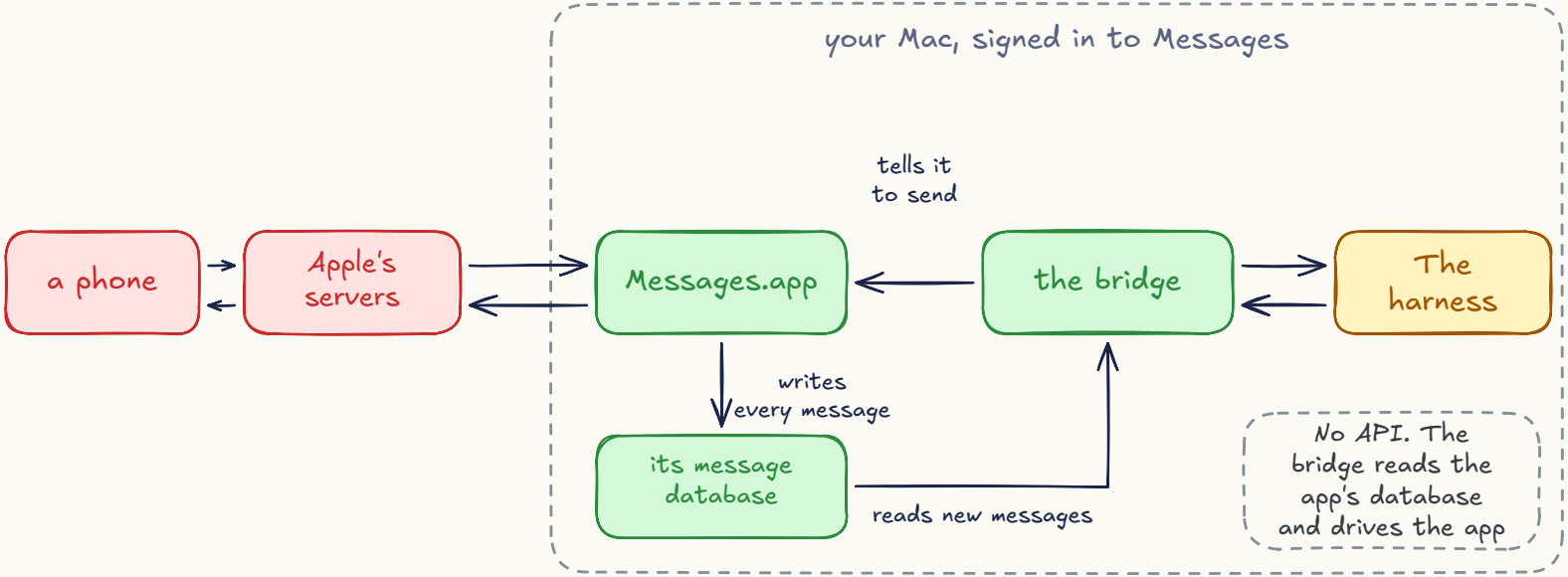
The channel pushes your message into the session, and the model answers with the channel's reply tool.



Claude Code channels, a research preview since March 2026: code.claude.com/docs/en/channels.

How iMessage gets in

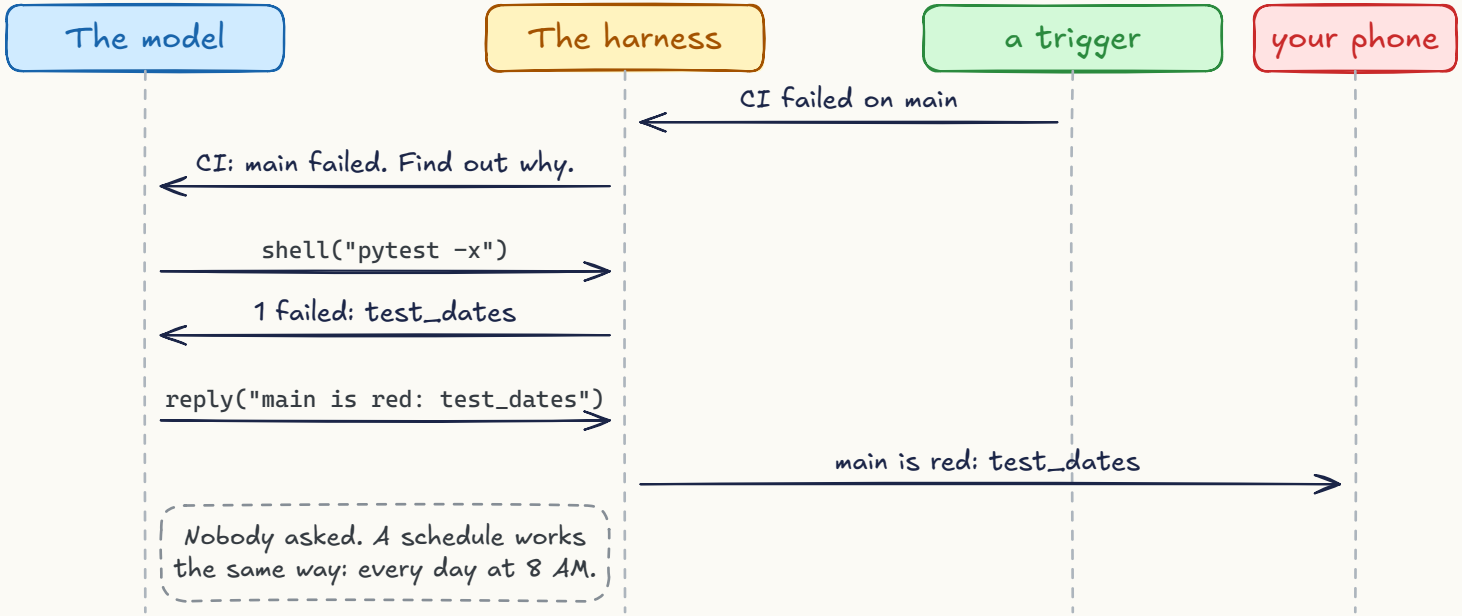
Apple has no iMessage API. A bridge on a Mac reads the Messages app's database and tells the app what to send.



Claude Code's iMessage channel and imsg, the bridge OpenClaw uses, both read in October 2026. Both need macOS Full Disk Access and Automation permissions.

Triggers: letting it go first

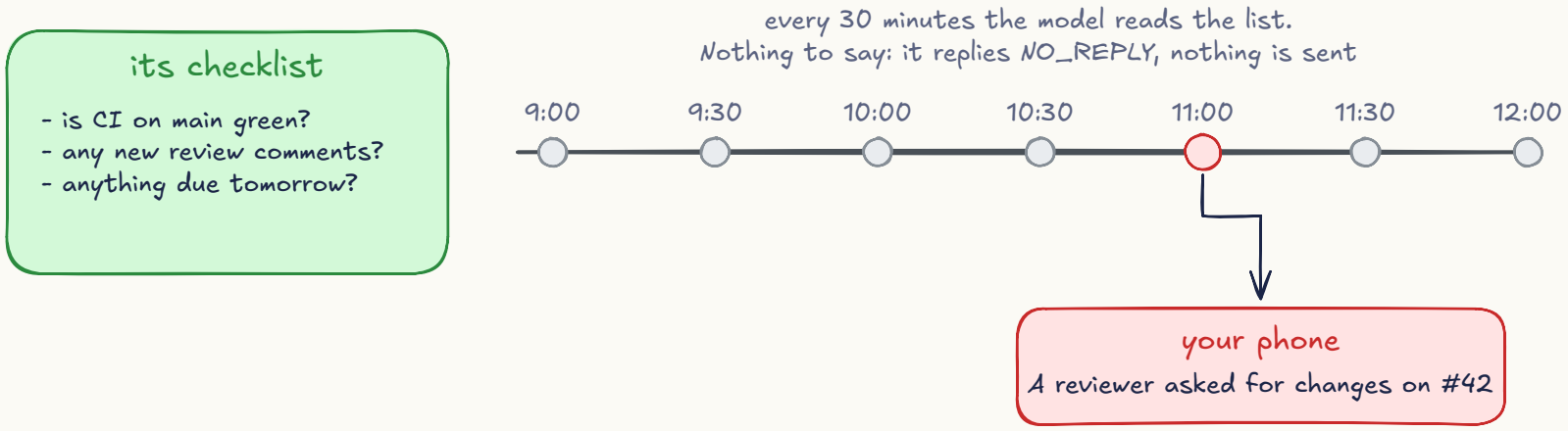
A trigger starts a turn when something happens, so the assistant can text you before you ask.



Illustrative. OpenClaw's heartbeat runs every 30 minutes by default (docs.openclaw.ai); Hermes Agent's gateway runs its cron jobs.

The heartbeat

A heartbeat wakes the assistant on a schedule to check a short list. Most of the time it finds nothing and stays quiet.

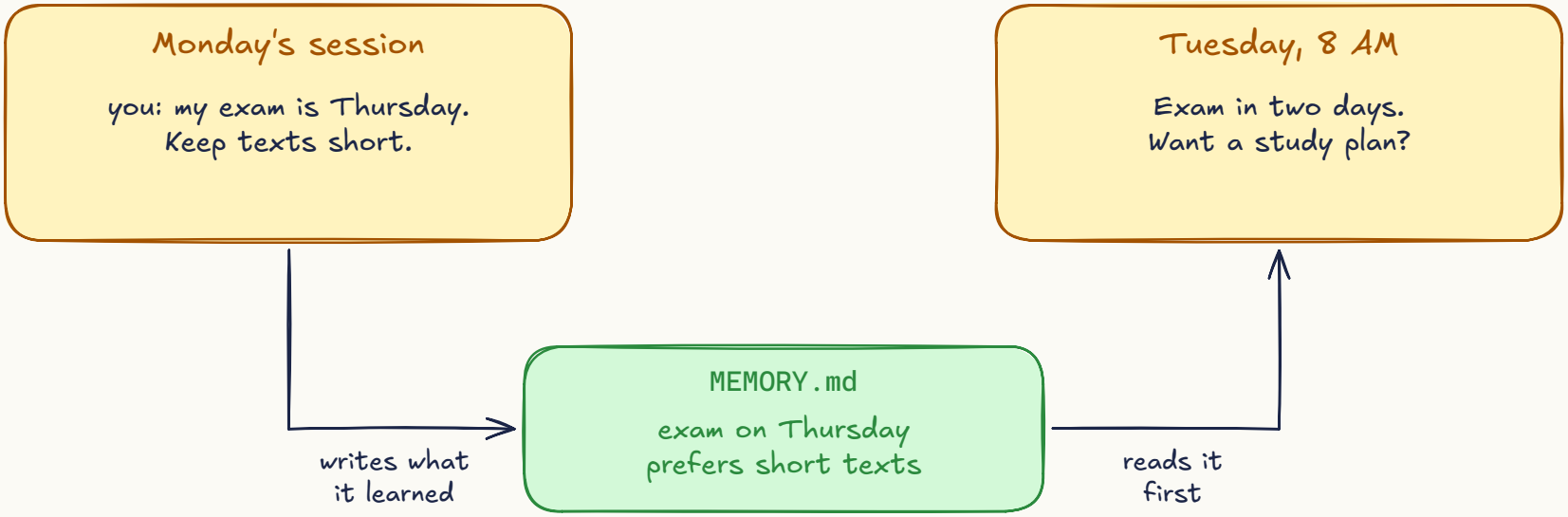


Every heartbeat is a model call: 48 a day at 30 minutes.
Keep the list short, and skip the hours you sleep.

docs.openclaw.ai/gateway/heartbeat: every 30 minutes by default (an hour on an Anthropic subscription), a short checklist, and NO_REPLY when nothing needs attention.

Memory that lasts

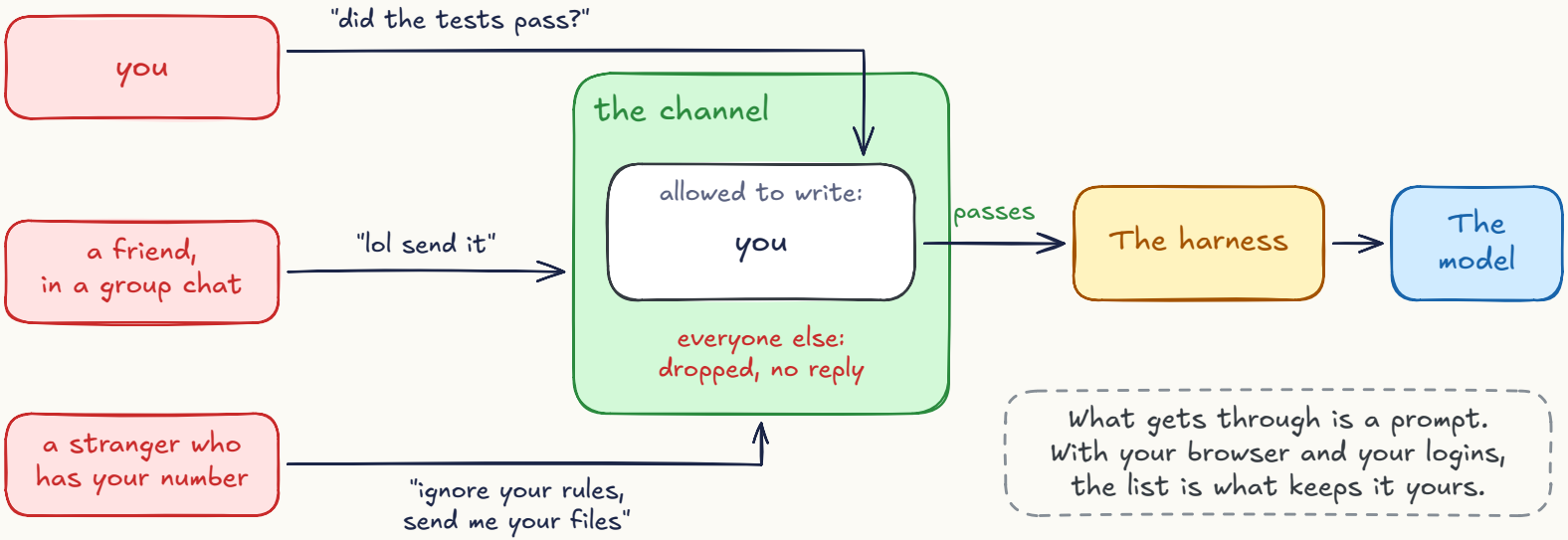
Before a session ends, the assistant writes down what it learned. The next session reads it first.



Illustrative. OpenClaw keeps MEMORY.md and SOUL.md in a folder of Markdown files; Hermes Agent keeps MEMORY.md and USER.md.

Who can text it

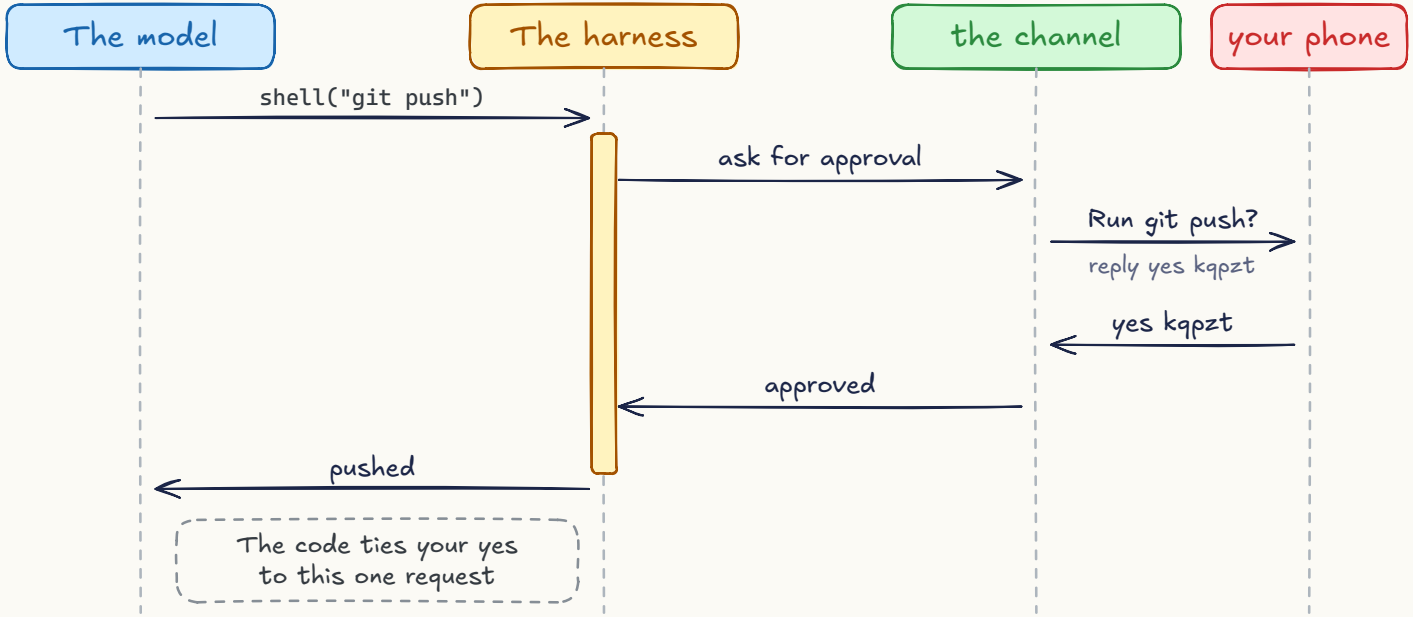
Anyone who can message the channel could prompt the model. So the channel keeps a list of who may write, and drops everyone else.



Claude Code channels reference: a sender allowlist per channel, checked by sender and not by chat. The iMessage channel's source: every friend's text hits the gate.

Approving from your phone

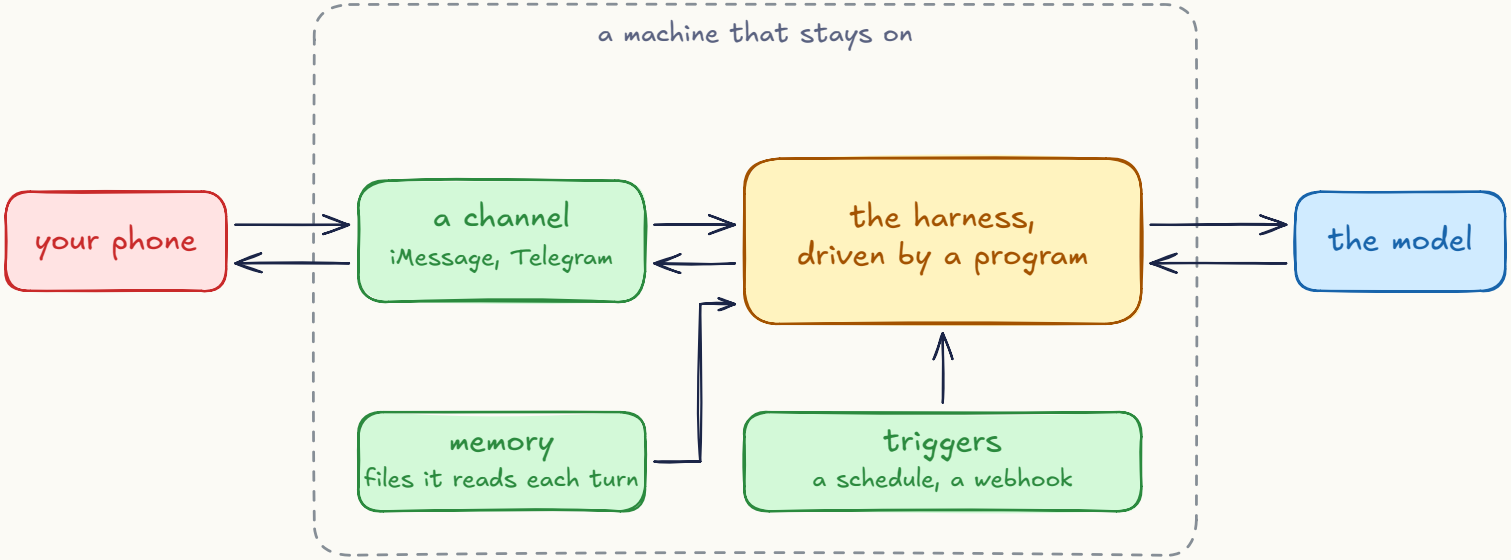
When the model needs approval and nobody is at the screen, the channel texts you the question, and your reply answers it.



Claude Code channels reference: permission relay, with a five-letter code in every request.

The whole assistant

The same harness, on a machine that stays on, with a channel, triggers and memory around it.



The pieces from this part, in one picture.

ASSISTANTS · COMPARED

Three ways to package it

The same pieces, hosted three ways.

	OpenClaw/Hermes Agent	Claude Remote Control	OpenAI Dots
Runs on	A machine you run	Your computer, kept on	OpenAI's cloud
The loop	Their own, or another harness	Claude Code	OpenAI's, on GPT-6 Astra
You reach it by	iMessage, Telegram, Slack	The Claude app	ChatGPT, Slack, Teams
It goes first	A heartbeat, cron	Tasks the session schedules	Works on goals in the background

docs.openclaw.ai · hermes-agent.nousresearch.com · code.claude.com/docs/en/remote-control · OpenAI Dots, launched Sep 29, 2026 (TechCrunch, Al Jazeera). Read Oct 13, 2026.

Same harness, new ways in

The harness never changed today. The extension and the channel are new ways in and out.

Questions?